

Nmap Network Scanning

Title: Host Discovery, Service Detection, OS Detection, and Port Scanning using Nmap

1.Introduction:

Objective

The objective of this project is to use the Nmap (Network Mapper) tool to perform network reconnaissance and security assessment tasks. The project focuses on:

- Discovering active hosts on a network
- Identifying open ports on target systems
- Detecting running services and their versions
- Identifying the operating system of the target machine
- Performing basic vulnerability detection using Nmap NSE scripts
- Understanding how attackers and security professionals gather network information

The project also aims to develop practical skills in cybersecurity scanning techniques and network analysis.

2.About Nmap:

What is Nmap?

Nmap (Network Mapper) is an open-source network scanning and security auditing tool used by system administrators and cybersecurity professionals. It helps identify live hosts, open ports, running services, operating systems, and potential vulnerabilities in a network.

Features of Nmap

- Host discovery
- Port scanning
- Service version detection
- OS fingerprinting
- Vulnerability scanning with NSE scripts
- Network inventory and monitoring

Why Nmap is Used

- Security auditing
- Network troubleshooting
- Vulnerability assessment
- Device inventory management
- Penetration testing preparation

3.Environment Setup:

Tools Used:

Tool	Purpose
Nmap	Network scanning and enumeration
Kali Linux / Ubuntu / Windows	Operating system used for scanning
Terminal / Command Prompt	Running Nmap commands

Target Information:

Target	Description
127.0.0.1 / Localhost	Local machine testing
Scanme.nmap.org	Public test server provided by Nmap

Note: Always perform scans only on authorized systems.

4. Host Discovery:

Purpose: Host discovery identifies active systems available on the network.

Command Used: nmap -sn 192.168.1.0/24

Command Explanation:

Option	Meaning
-sn	Ping scan without port scanning
192.168.1.0/24	Target subnet

Screenshot:

```
(sanika@kali)-[~]
$ nmap -sn 192.168.61.142/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-17 04:56 EDT
Nmap scan report for 192.168.61.1
Host is up (0.0018s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.61.2
Host is up (0.00053s latency).
MAC Address: 00:50:56:FD:EB:39 (VMware)
Nmap scan report for 192.168.61.142
Host is up (0.0013s latency).
MAC Address: 00:0C:29:6E:C4:43 (VMware)
Nmap scan report for 192.168.61.254
Host is up (0.0035s latency).
MAC Address: 00:50:56:F8:B8:50 (VMware)
Nmap scan report for 192.168.61.143
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.22 seconds
```

Observation:

- Active hosts detected in the network.
- IP addresses and MAC addresses identified.

5. Port Scanning:

Purpose: Port scanning identifies open ports on the target system.

Command Used: nmap 192.168.1.1

Command Explanation:

Option	Meaning
nmap	Starts scan
192.168.1.1	Target IP address

Screenshot:

```
(sanika@kali)-[~]
$ nmap 192.168.61.142
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-17 05:02 EDT
Nmap scan report for 192.168.61.142
Host is up (0.0015s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:6E:C4:43 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.17 seconds
```

Observation:

Port	State	Service
22	Open	SSH
80	Open	HTTP
443	Open	HTTPS

6. Service Version Detection:

Purpose: Service detection identifies application versions running on open ports.

Command Used: `nmap -sV 192.168.1.1`

Command Explanation

Option	Meaning
-sV	Detect service versions

Screenshot:

```
(sanika@kali)~$ nmap -sV 192.168.01.142
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-17 05:03 EDT
Nmap scan report for 192.168.01.142
Host is up (0.013s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:6E:C4:43 (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.13 seconds
```

Observation:

Port	Service	Version
22	OpenSSH	OpenSSH 8.2
80	Apache HTTPD	Apache 2.4.41

7. Operating System Detection:

Purpose: OS detection attempts to identify the target operating system.

Command Used: `nmap -O 192.168.1.1`

Command Explanation

Option	Meaning
-O	OS detection
sudo	Administrative privileges required

Screenshot:

```
(sanika@kali)-[~]
$ nmap -O 192.168.61.142
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-17 05:03 EDT
Nmap scan report for 192.168.61.142
Host is up (0.0020s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:6E:C4:43 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.30 seconds
```

Observation:

- Target OS detected as Linux.
- Kernel information partially identified.

8. Aggressive Scan:

Purpose: Aggressive scan performs OS detection, version detection, script scanning, and traceroute.

Command Used: nmap -A 192.168.1.1

Command Explanation:

Option	Meaning
-A	Aggressive scan

Screenshot:

```
(sanika@kali):~$ nmap -A 192.168.61.142
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-17 05:04 EDT
Nmap scan report for 192.168.61.142
Host is up (0.0022s latency).
Not shown: 277 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ ftp-syst:
|_ STAT:
|_ FTP server status:
|_   Connected to 192.168.61.142
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   vsFTPd 2.3.4 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ ssh-hostkey:
|_   1024 60:0f:cf:6a:7a:d6:98:2a:fa:ck:d5:6c:cd (DSA)
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:eb:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ sslv2:
|_   SSLv2 supported
|_   ciphers:
|_   SSL2_DES_64_CBC_WITH_MD5
|_   SSL2_DES_192_CBC3_CBC_WITH_MD5
|_   SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_   SSL2_RC4_128_WITH_MD5
|_   SSL2_RC2_128_CBC_WITH_MD5
|_   SSL2_RC4_128_EXPORT40_WITH_MD5
|_ ssl-cert: Subject: commonName=ubuntu004-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-10T14:07:45
|_ smtp-command: netasploit@019.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ ssl-date: 2026-05-17T09:05:14+00:00; +7s from scanner time.
53/tcp    open  domain       ISC BIND 9.4.2
|_ dns-nsid:
|_   bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_ http-title: Netasploit@02 - Linux
```

Observation:

- Detailed service information collected.
- Additional scripts executed.
- Traceroute information displayed.

9. Vulnerability Scanning:

Purpose: Nmap NSE (Nmap Scripting Engine) scripts can identify common vulnerabilities.

Command Used: nmap --script vuln 192.168.1.1

Command Explanation:

Option	Meaning
--script vuln	Runs vulnerability detection scripts

Screenshot:

```
(saaike@kali)~$ nmap --script vuln 192.168.0.142
Starting Nmap 7.95 ( https://nmap.org ) at 2020-05-17 05:17 EDT
Nmap scan report for 192.168.0.142
Host is up (0.0045s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|     vsFTPD version 2.3.4 backdoor
|       State: VULNERABLE (Exploitable)
|       IDS: BID:48539 CVE:CVE-2011-2523
|       vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|       Disclosure date: 2011-07-03
|       Exploit results:
|         Shell command: id
|         Results: uid=0(root) gid=0(root)
|       References:
|         https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|         https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|         https://www.securityfocus.com/bid/48539
|         http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|
| 22/tcp    open  ssh
| 23/tcp    open  telnet
| 25/tcp    open  smtp
| ssl-dh-params:
|   VULNERABLE:
|     Anonymous Diffie-Hellman Key Exchange MitM Vulnerability
|       State: VULNERABLE
|       Transport Layer Security (TLS) services that use anonymous
|       Diffie-Hellman key exchange only provide protection against passive
|       eavesdropping, and are vulnerable to active man-in-the-middle attacks
|       which could completely compromise the confidentiality and integrity
|       of any data exchanged over the resulting session.
|       Check results:
|         ANONYMOUS DH GROUP 1
|           Cipher Suite: TLS_DH_anon_WITH_RC4_128_MD5
|           Modulus Type: Safe prime
|           Modulus Source: postfix builtin
|           Modulus Length: 1024
|           Generator Length: 8
|           Public Key Length: 1024
|       References:
|         https://www.ietf.org/rfc/rfc2246.txt
|
| Transport Layer Security (TLS) Protocol DHE_EXPORT Ciphers Downgrade MitM (Logjam)
|   State: VULNERABLE
|   IDS: BID:74733 CVE:CVE-2015-4000
|   The Transport Layer Security (TLS) protocol contains a flaw that is
|   triggered when handling Diffie-Hellman key exchanges defined with
|   the DHE_EXPORT cipher. This may allow a man-in-the-middle attacker
|   to downgrade the security of a TLS session to 512-bit export-grade
|   cryptography, which is significantly weaker, allowing the attacker
|   to more easily break the encryption and monitor or tamper with
|   the encrypted stream.
|   Disclosure date: 2015-5-19
|   Check results:
|     EXPORT-GRADE DH GROUP 1
|       Cipher Suite: TLS_DHE_RSA_EXPORT_WITH_DES40_CBC_SHA
|       Modulus Type: Safe prime
|       Modulus Source: Unknown/Custom-generated
|       Modulus Length: 512
|       Generator Length: 8
|       Public Key Length: 512
|   References:
|     https://weakdh.org
|     https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2015-4000
|     https://www.securityfocus.com/bid/74733
```

Observation

- Potential vulnerabilities identified.
- Services with outdated versions highlighted.

10.Common Nmap Scan Types:

Scan Type	Command	Purpose
Ping Scan	<code>nmap -sn target</code>	Discover live hosts
TCP SYN Scan	<code>nmap -sS target</code>	Stealth port scanning
UDP Scan	<code>nmap -sU target</code>	Scan UDP ports
Service Detection	<code>nmap -sV target</code>	Detect versions
OS Detection	<code>nmap -O target</code>	Identify operating system
Aggressive Scan	<code>nmap -A target</code>	Detailed scan

11. Analysis:

Scan Analysis

Host Discovery Analysis: The host discovery scan successfully identified active devices available in the network range. This step helped determine which systems were reachable and responding to network requests.

Port Scanning Analysis: The port scan revealed several open ports on the target system. Open ports indicate services actively listening for connections. Common ports such as:

- Port 22 → SSH service
- Port 80 → HTTP service
- Port 443 → HTTPS service

were identified during scanning.

Service Detection Analysis: Service version detection provided detailed information about applications running on open ports. This information is useful because outdated software versions may contain known vulnerabilities.

Example:

- Apache HTTP Server version
- OpenSSH version
- FTP or database services

OS Detection Analysis: Operating system detection identified the probable operating system of the target machine. Nmap compares TCP/IP fingerprints against its database to estimate the OS.

Vulnerability Scan Analysis: The NSE vulnerability scripts highlighted possible weaknesses in running services. Vulnerabilities may occur due to:

- Outdated software
- Weak configurations
- Exposed unnecessary services
- Misconfigured servers

The scan demonstrated how Nmap can assist security professionals in identifying risks before attackers exploit them.

12. Conclusion:

The Nmap scanning project successfully demonstrated the use of network reconnaissance and security auditing techniques. Through different scanning methods, active hosts, open ports, running services, and operating systems were identified effectively.

The project provided practical understanding of:

- Host discovery techniques
- Port and service enumeration
- Operating system fingerprinting
- Basic vulnerability assessment
- Importance of network security monitoring

The analysis showed that exposed services and outdated software may create security risks if not properly managed. Regular scanning and vulnerability assessments are important for maintaining secure systems and networks.

Overall, Nmap proved to be a powerful and flexible tool for cybersecurity professionals, penetration testers, and network administrators.